

OSCN Template

1. What makes an OS?

The role of an operating system serves two main purposes: the management of system resources such as CPU time and storage, and to act as a bridge between the user and applications. Modern operating systems are composed of five elements:

1. A kernel, which is an essential program which links applications to hardware.
2. A file system, which is responsible for how data is stored on LTSD's.
3. A programming language (such as C, Holy-C for TempleOS, or Assembly)
4. A user interface, GUI or CLI.
5. An API, or App Program Interface which allows for communication between programs.

2. What is virtual memory and its purpose?

System operational memory is physically limited and at times is not enough for application demands. For this reason the OS (which is responsible for resource management) allocated pages to 'cold' data stored in operational memory to memory found in a LTSD. Although this memory is not as fast and efficient as operational memory, it prevents tasks from crashing. Although definitely not a large amount of LTSD memory should be used for virtual memory, and the recommended ratio is 3:1.

3. Advantages/Disadvantages of a CLI

There are two main types of user interfaces: GUI (Graphical User Interface) and CLI (Command Line Interface). The only graphical element of a CLI is text alone. This allows CLI's to weigh very little overall in terms of occupied storage and removes bandwidth limits. Additionally, CLI's provide faster and deeper control over your operating system with the correct command knowledge. However, the commands and how they are used are not ideal for inexperienced users and require highly skilled and expensive personnel.

4. Advantages/Disadvantages of batch processing

Batch processing is a computational model which involves an end user inputting data which is then sorted into 'batches. These batches are fed into a powerful mainframe computer which handles these batches one-by-one, once at a time. The advantages include the effective use of physical resources which are dedicated to one batch, and how users are forced to write correctly due to the time an output can typically take. Although this could also be of huge inconvenience to these users and another issue is the lack of human interaction with the system itself.

5. What is a server?

A common misconception is that a server is a dedicated powerful computer. But in reality, a server is simply a role a device takes to provide services to end users. The reason it usually is assigned to a powerful mainframe is because other devices cannot handle many requests due to their operating system and physical limits, as well as cost effectiveness. More specifically, the devices typically assigned with this role are very powerful computers capable of running many calculations, the output of which is stored on dedicated storage arrays.

6. What are private IP's, why were they introduced?

Regular IP addresses are used to describe the approximate location from which a connection is being made, and they are assigned by your ISP whose code occupy the address' first bits. But these addresses are not limitless, and to solve this scarcity private IP's were introduced. Private IP's can be used by multiple devices unlike regular IP's, but within the same isolated local network. This allows for communication between devices connected to the same network. For communication beyond this network, however, this requires a NAT (Network Address Translation) which converts this IP to a public one visible elsewhere.

7. What is cloud computing, why is it used?

There is a saying - "there is no cloud, it's just someone else's computer". Cloud computing is a service provided to clients with huge data which offers them a convenient, redundant and safe way to store it. There are three types of cloud storage: private, hybrid and public, all of which determine where this data is stored depending on the needs of the client (i.e. GDPR protected data vs. a web application). The types of these cloud services come in three forms: PaaS (Platform as a service), IaaS (Infrastructure as a service) and SaaS (Software as a Service).

8. Wireless tech: advantages/disadvantages

Wireless technologies' most important advantage is increased range with relatively easier installation. It is with thanks to them that portable devices have become so popular. They do not require cables which could be expensive to install and may require a lot of care. In fact, wireless alternatives are much faster to install. They are also relatively cost effective long-term even if they have higher initial costs.

Some disadvantages include security which is often an issue with poor encryption or password protection. Wireless technologies are significantly slower in data transfer speeds compared to wired alternatives. The farther the user is from the signal origin, the worse it is. It is also prone to interference most especially in shared bands which are the most widely used because they are free.

9. Cooperative and Pre-emptive multitasking

Computers come with limited resources, ranging from CPU time to RAM operational memory and physical storage. Yet despite these limitations, opening multiple applications, with often higher requirements, is still possible with thanks to cooperative and pre-emptive multitasking.

Before, resources were allocated by applications to themselves to meet requirements, and each app relied on cooperation where each would no longer use resources they needed.

But this caused several issues mainly because of incorrectly written apps.

Today, most operating systems hold the role of assigning computational resources to applications. If applications misbehave, the OS can shut them down and can give these resources to another.